StoryTutor-MM

Product Plan — Analysis & Execution Brief

Objective
Turn the ASU-Sol research prototype into a cloud-hosted, multi-tenant tutor that Indian schools can pay for, serving Classes 6–8 with approved curriculum content in English, Hindi and Marathi.

Core product promise

- Answers only from the approved textbook and displays the exact page.
- Explains concepts at the learner’s level and in the learner’s language.
- Reads answers aloud and generates a fact-checked lesson video of about 30 seconds.
- Schools do not buy GPUs; shared cloud infrastructure handles the heavy compute.

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Prepared from the supplied PRODUCT_PLAN.md. This PDF organizes and analyzes the source plan; technical, licensing and cost statements are presented as claims from that plan, not as independently verified facts.

1. Executive Summary

The plan is not primarily a feature-building exercise. It is a transition from a working research prototype to a dependable school SaaS product. The key work is to remove legal exposure, establish school-level access isolation, move off borrowed ASU compute, and build the teacher-facing workflows that make schools willing to pay.

Area	Current position in source	Business implication
Core learning experience	Grounded chat, page citations, hybrid retrieval, guardrails, TTS and video pipeline marked working.	A credible prototype exists; preserve and validate it rather than rewriting everything.
Commercial readiness	Auth, tenancy and per-school data isolation absent.	Not ready for real school/student accounts.
Legal readiness	TTS model-weight licensing and NCERT permissions unresolved.	A launch blocker, not a later paperwork task.
Hosting	LLM runs on borrowed ASU Sol compute; target is managed shared cloud.	Deployment and cost reliability need to be proven away from ASU.
School adoption	Teacher dashboard, learner memory and progress reporting absent.	The product needs teacher value, not only a student chatbot.

Recommended sequence: legal clearance and P0 hardening → controlled two-school pilot → measure learning outcomes and operating costs → convert pilots into paid contracts → scale infrastructure and features.

2. What Exists Today

Capability	Status in source	Notes
Grounded chat with [S1] citations and SSE streaming	Working	rag_chat.py; static/chat.html
Dense + BM25 retrieval, RRF k=60, optional reranking	Working	rag_engine.py, bm25.py, reranker.py
Evidence gate, one rewrite rescue, refusal	Working	Minimum evidence score stated as 0.25
Injection/PII/citation/groundedness guardrails	Working	guardrails.py
Trilingual TTS	Working	Parler → MMS fallback; licensing must be resolved
Video generation with images, narration, subtitles and per-claim fact-check	Working	Pipeline in story_mvp/video/
268-question evaluation harness	Built	Reports currently live only on Sol
Auth, tenancy, per-school data	Absent	Must be built before commercial use
Teacher dashboard, learner memory, progress reports	Absent	Important for school adoption
output_type / difficulty controls	Not honoured	Values are transported but not applied

Data consistency and immediate bugs

- Use 7,086 chunks as the single corpus count; the source says other documents contain conflicting figures.
- Fix /api/options, which the source says will return HTTP 500 due to unimported constants.
- Remove unresolved Git merge-conflict markers from README.md.
- Replace the hardcoded Flask SECRET_KEY with a secure environment-managed secret.

3. Legal, Licensing and Child Privacy

The source plan explicitly places licensing and permissions before commercial product code. This is the most important launch gate because a technically functional demo can still be unusable in a paid product.

Asset / issue	Concern in source	Action
MMS Hindi/Marathi/English TTS	CC-BY-NC-4.0 is non-commercial.	Do not ship in paid product unless a valid commercial license is obtained; replace it or keep only in a clearly disclosed research environment.
Indic Parler TTS	Code documents Apache-2.0, but model weights may have separate terms.	Verify the exact weight license and get written confirmation. Replace or negotiate if commercial use is not allowed.
NCERT PDFs	Commercial extraction and serving may require permission.	Seek NCERT/district authorization; build ingestion for district-provided corpora; use an agreed pilot/MOU until rights are clear.
DPDP Act 2023 / children's data	Plan calls out parental consent, no behavioural profiling and no targeted ads.	Document privacy, consent, retention, deletion, access controls and deployment-region practices; obtain appropriate legal review.

Privacy claims must be accurate

The source proposes a strong sales message around Indian data residency, no model training on student questions, and no tracking. These statements should only be used after the deployment, vendor contracts, telemetry and data flows prove them. Avoid absolute promises until each is validated.

Practical launch checklist

- Create a model-and-dataset license register with exact versions, links, owners and evidence of commercial rights.
- Obtain written permission or use school/district-supplied content with documented rights.
- Write a parent-facing privacy one-pager in plain language.
- Define parental consent, data minimization, retention/deletion and incident-response procedures.
- Review all vendors and hosting regions before claiming that data stays in India.

This section is a product-risk analysis, not legal advice. Licensing and DPDP obligations should be reviewed by qualified counsel.

4. Target Architecture and Cost Model

The plan recommends a hosted multi-tenant SaaS: a shared application tier, object storage, a dedicated interactive GPU path, and separate asynchronous media workers. This separation is sensible because video generation can consume compute for minutes while chat needs low latency.

Layer	Proposed role	Design note
Application tier	Flask app, authentication, tenant routing, streaming chat	Two containers are proposed; enforce tenant isolation at every data-access boundary.
Object storage	Videos, indexes and caches	Use tenant-scoped permissions and lifecycle/retention rules.
Interactive GPU	L4/A10G 24 GB; vLLM Qwen3-8B; bge-m3; optional reranker	Keep the chat path responsive; load test concurrent sessions before committing to sizing.
Media workers	Spot/preemptible GPU for image generation, TTS and ffmpeg	Use a durable queue, retries, idempotent jobs and job-status reporting.
Operations	Health checks, metrics, backups, metering	Add tenant-level latency, groundedness, refusal and cost visibility.

Cost assumptions in the source

- Interactive node: approximately $0.6–0.9/hour, stated as roughly ■15k–18k/month.

- Batched media: approximately ■5k–10k/month at pilot volume.
- Estimated infrastructure: ■15–25 per student/month under the stated 20-school / 300-students-per-school sharing scenario.

Treat these as planning estimates, not a quote. Validate actual provider prices, GPU uptime, utilisation, storage, bandwidth, observability, support, taxes and peak usage. Run a load test and calculate cost per active student and per generated video.

5. Prioritized Execution Roadmap

P0 — Commercial blockers (source estimate: 2–3 weeks)

Priority	Deliverable	Acceptance evidence
P0.1	Resolve TTS/model licenses and textbook permission path.	Written commercial rights or a documented replacement; corpus rights recorded.
P0.2	Authentication and multi-tenancy: school accounts, student class codes, tenant-specific libraries/storage/rate limits.	Automated tests prove one school cannot access another school's users, files, videos or usage data.
P0.3	Fix /api/options, README merge markers and hardcoded secret; quarantine legacy /api/generate.	Clean startup, passing tests, secrets loaded from environment.
P0.4	Versioned prebuilt index and readiness endpoint.	Service reports ready only after required index/model dependencies are available.
P0.5	Dockerize app, LLM, TTS and media worker.	One-command reproducible deployment outside ASU; documented configuration and rollback.

P1 — School value (source estimate: about 1 month)

Priority	Deliverable	Why it matters
P1.1	Teacher surface: assign chapter, view common misconceptions, print quiz, view learner library.	Teachers are the adoption path and need actionable classroom value.
P1.2	Quality-tier dashboard by tenant.	Makes model/provider changes and quality degradation visible.
P1.3	Consent-based learner memory for quiz misses and progress.	Supports personalization and progress reports while requiring careful child-data controls.
P1.4	Commit evaluation reports and run 53 chat questions nightly per environment.	Makes quality regressions visible before schools report them.

P2 — Scale and polish

- Move from IndexFlatIP to IVF/HNSW or a prebuilt FAISS artifact when corpus growth makes it worthwhile; source suggests revisiting beyond 50k chunks.
- Add per-request observability, backups and per-tenant usage metering.
- Implement actual output_type and difficulty behaviour.

6. Pilot and Go-to-Market

The proposed go-to-market is a free one-term pilot with two schools: one Marathi-medium and one English-medium. The pilot should establish learning value and operational reliability, then convert to paid contracts using evidence gathered on the schools' approved content.

Pilot workstream	Measure / evidence to collect
Learning impact	Pre/post assessment results, completion and retention where appropriate, with a defensible comparison design.
Answer quality	Citation correctness, answer groundedness, refusal rate, hallucination/error reports and teacher review.
Language quality	Teacher/student feedback on English, Hindi and Marathi explanations and pronunciation.
Product adoption	Weekly active learners, teacher activity, repeat usage and assignment/quiz usage; avoid invasive behavioural profiling.
Reliability	First-token latency, video completion time, failures, queue wait and service uptime.

Pilot workstream	Measure / evidence to collect
Economics	GPU hours, cost per active student, cost per video and support effort per school.

Positioning from the source plan

> **Answers only from your book · shows the page · works in Marathi · nothing leaves India.**

Use this only after validating the claims. A district-first pathway and ingestion for district-provided content can reduce dependency on a single textbook corpus and support a clearer enterprise sales conversation.

7. Main Risks and Mitigations

Risk	Mitigation from / interpretation of source
NCERT permission unavailable	Prioritize district-provided content ingestion and a documented 'bring your approved textbook' workflow.
No commercially usable Hindi/Marathi TTS	Evaluate alternatives now; treat voice as a core promise and test quality before choosing.
One founder carries infra and sales	Use managed services where appropriate, minimize P0 scope, document deployment and incident response.
Schools will not pay without evidence	Design the pilot to measure learning outcomes and teacher value from day one.
Internet outages disrupt lessons	Plan cached content and queued sync as a later hybrid/offline tier; define what works offline explicitly.
GPU costs or latency exceed assumptions	Load-test realistic concurrency, queue media separately, measure cost per student/video and set usage limits.
Child data crosses tenant boundaries	Threat-model tenant isolation and add automated negative-access tests before pilot.

8. Definition of Done

- A Class 7 learner can ask in Marathi on a school Chromebook or phone and receive a grounded answer with a clickable page citation; first token is under one second on the product’s own cloud, not ASU.
- Commercial licensing and textbook rights are cleared; DPDP posture is documented and a parent-facing privacy one-pager exists.
- A teacher can log in, inspect class questions and print a quiz without IT assistance.
- A visible quality/health card reports which model tier is serving, so degradation is not silent.
- Economics work at 20 or more schools on one interactive node, with the media queue draining on spot capacity.

9. Decisions to Make Before Launch

Decision	Recommended next move
Content rights	Choose: licensed NCERT corpus, authorized district content, or a clearly scoped pilot with written permission.
TTS engine	Verify exact model-weight terms and benchmark naturalness, latency and Hindi/Marathi coverage.
Cloud provider	Choose using Indian region availability, GPU price, data-processing terms, service reliability and operational simplicity.
School data model	Define school, teacher, class, learner, content and job ownership before implementing tenancy.
Pilot success threshold	Agree on learning, quality, reliability, adoption and cost targets before starting the pilot.
Pricing	Treat ■15–25/student/month as an infrastructure estimate, not selling price; add support, taxes, margin and school procurement realities.

10. Recommended First 10 Working Days

Day	Focus	Output
1	Freeze product claims; inventory every model and corpus license.	License register and list of unresolved legal questions.
2	Fix /api/options, README conflicts and secret handling.	Clean baseline branch and passing tests.
3–4	Design tenant data model and threat model.	Schema, access rules and isolation test plan.
5–6	Implement authentication, school/class access and tenant-scoped storage.	Working login and automated cross-tenant denial tests.
7	Version index build and health/readiness checks.	Reproducible index artifact and readiness endpoint.
8	Dockerize the application services.	Local reproducible deployment.
9	Deploy to candidate cloud and run chat/video smoke tests.	Deployment notes and latency/cost measurements.
10	Review pilot protocol with prospective schools/teachers.	Pilot agreement outline and measurement plan.

Final assessment

StoryTutor-MM appears to have a meaningful technical base according to the supplied inventory, but it is not yet commercially ready. The highest-leverage move is disciplined sequencing: clear rights, protect children and school data, make deployment reproducible, then prove teacher and learner value through a measured pilot. Avoid adding advanced features until the P0 launch gates are closed.

Source: supplied PRODUCT_PLAN.md (dated 18 October 2026). This report reorganizes and analyzes that source. It does not independently verify licensing, legal interpretations, vendor prices, model performance or technical claims.